

sector. Freescale has also developed a low-cost intelligent hospital kiosk, which is a biometric data acquisition system that assesses a person's health and performs basic medical tests for body temperature, blood pressure, etc. After checking the health parameters, the portable USB relays the biometric data over the Internet to a hospital or physician for quick investigation and treatment.

Made in India MPC830x launched.

Further, Freescale Semiconductor launched a new portfolio of cost-effective PowerQUICC II Pro processors with exceptional performance-to-power ratios and upgraded interconnect technology options. Featuring sub-\$10 resale price points, the new MPC830x communications processor portfolio, which includes the MPC8308, MPC8309 and MPC8306/8306S products, offers upgraded high-speed interconnects allowing customers to lower bill-of-material (BOM) costs for target applications such as line cards, industrial applications, WLAN access points, intelligent NIC equipment, smart energy gateways and general-purpose control functions. The MPC830x was developed from concept to production at the India design centre.

The Industrial MCU revolution.

Embarking on a new era of microcontroller (MCU) leadership, a session was conducted on Kinetis family of 90-nanometre (nm) 32-bit MCUs based on the new ARM Cortex-M4 processor. Complementing Freescale's recently announced 90nm ColdFire+ MCU line, Kinetis represents one of the most scalable portfolios of low-power, mixed-sig-

nal ARM Cortex-M4 processor-based MCUs in the industry.

Customers walk the Freescale path!

Alongside, Freescale has enabled a large number of customers in India like Gill Instruments, Magphy Systems, Evolute Systems and Bisquare Systems to build innovative products using Freescale technology. These customers also showcased their products at *FTF 2010*. For instance, the level gauge system displayed by Gill Instruments used Freescale technology to measure water levels and transmit it to the base unit that can be installed 500 km away from the tank. The base unit can thereafter prompt action for the pump. The unit also generates alerts on leakage or dry run when no water is being pumped.

"We hope to build a single base unit monitoring eight different tanks and connect it to the pump. We are also working on modifying this system to measure fuel tank levels in automobiles," says Gurjeet Singh, director, Gill Instruments.

Bisquare System, on the other hand, has built an LED-based solar chargeable lamp called EccoDiva for the rural sector in mind, costing Rs 1475. "Freescale technology helped us develop the EccoDiva controller which makes the lamp highly efficient. As a result, our product on power-backup side is 94-95 per cent efficient compared to other solar chargers which are only 80 per cent efficient," says Hemendra Varshney, manager, electronics, Bisquare Systems.

Another innovative product was Impress, a wireless mobile thermal printer by Evolute Systems that has a two-track magnetic card reader, single or dual smart card readers, fingerprint scanner, bar code reader interface, etc integrated into it. Magphy's high-frequency inverters designed using Freescale's digital signal controllers too claimed technological superiority over the traditional 50Hz transformer inverters.

Software vendors—strong comrades

Apart from customers and partners, Freescale has software platform vendors supporting its innovation *mantra*



Dr Reza Kazerounian-senior vice president and general manager, microcontroller solutions, Freescale Semiconductor, presenting keynote address during FTF 2010

and the Great Electronics Dream. They displayed their multi-core capabilities. ENEA, for instance, highlighted ENEA Optima, while Mentor Graphics showcased its Nucleus RTOS solutions, multi OS on multicore solutions. "We are trying to show comprehensive support for Freescale processors from power architecture to Starcore DSPs," says Magnus Gille, director of product enablement, ENEA Software.

QNX, on the other hand, demonstrated a QNX car application platform that allows development teams to create infotainment systems that include Internet connectivity, Bluetooth, rear-seat gaming, vehicle monitoring, navigation, etc. Wind River too showcased an on-chip debugging solution and system simulation with Wind River Simics.

The spectacle of the year

In a nutshell, as Kalpesh Gala, director business development, Mentor Graphics, rightly stated, "*FTF India* is a spectacular event to learn the Indian customers' needs and opportunities." Attended by 1400 professionals, featuring over 70 hours of technical training sessions and over 50 interactive technology lab demonstrations, *FTF India 2010* showed immense promise for the Indian companies including enterprises, small and medium enterprises (SMEs) and public-sector undertakings (PSUs). Alongside, it presented an apt platform for the design community to explore the latest innovations in embedded design and exchange ideas with industry experts. ■

Home Energy Gateway!

Freescale demonstrated a comprehensively integrated home energy gateway solution that included appliance control through Zigbee protocol, facilitation of energy statistics information to the utility through the energy gateway and home security features along with automation features integrated for touch based programming as well as remote management through the internet cloud. Well orchestrated demonstration of the same was the highlight of the Tech lab at *FTF 2010* that had over 50 demonstrations.